Real-Time Vehicle Detection in Satellite Videos: Transitioning From Large Scenes to Clusters

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***Abstract*— Detecting vehicles from satellite videos presents several significant challenges: 1) satellite video frames typically possess extremely high resolutions, often containing millions or even hundreds of millions of pixels, whereas onboard computational resources remain constrained; 2) the nonuniform spatial distribution of vehicles results in inefficient allocation of computational resources; and 3) vehicles are typically small with limited distinguishing features, further complicating the detection task. In this article, we propose large scenes to clusters (LSCNet), an efficient and lightweight satellite video vehicle detection network specifically designed to address these challenges. To alleviate the difficulties associated with large-scale imagery and uneven vehicle distributions, we introduce a plug-and-play object cluster module (OCM). The OCM leverages interframe information from satellite video to adaptively identify and prioritize clustered regions, thereby enhancing detection precision. Furthermore, to improve the extraction of discriminative features from small-sized vehicles, we propose a lightweight multiframe feature aggregation module (MFAG), which effectively captures the spatiotemporal characteristics of vehicles while maintaining computationally efficiency. In addition, we refine the regression loss function by integrating the Kullback–Leibler divergence (KLD), enabling the generation of high-quality bounding boxes and significantly boosting the detection performance for small objects. Experimental evaluations on the Jilin-1 satellite video dataset demonstrate that the proposed method achieves improved detection accuracy while maintaining real-time performance, thereby validating its robustness and practical effectiveness.**

***Index Terms*— Feature aggregation, nonuniform object distribution, real-time detection, satellite video, small object.**

I. Introduction

RECENTLY, video satellites have emerged as a transformative advancement in Earth observation technology, capable of capturing continuous video sequences of designated regions, thereby enabling sustained surveillance of specific areas. These satellites exhibit considerable application potential in domains such as urban monitoring, marine object tracking, and traffic flow management [1], [2], [3]. In addition, the increasing demand for real-time surveillance of strategic regions and global hotspots has positioned moving object detection in satellite video as a pivotal area in the advancement of remote sensing satellite technology [4], [5]. By facilitating the real-time identification of high-interest targets, video satellites significantly enhance urban governance, optimize resource allocation, and strengthen security and early-warning systems.



Fig. 1. Challenges for vehicle detection in satellite video. (a) Large-scale scenes and the nonuniform distribution of vehicles. (b) Small vehicle size with limited features. (c) Enlarged portion of (b).

SkySat and Jilin-1 are among the most widely used satellite video datasets in contemporary research [6]. Satellite videos provide a broad field of view from an aerial perspective, exhibiting distinctive characteristics compared to natural images. However, object detection in satellite videos presents three primary challenges [7], [8], [9]. First, satellite videos cover a wide imaging range and contain a high pixel density, necessitating blockwise processing to improve computational efficiency. Second, vehicles in satellite videos are unevenly distributed, primarily aggregating along urban roads and bridges, while most pixels in background regions contribute minimally to the detection task. Finally, vehicles occupy a very small fraction of the video frame, frequently consisting of only a few dozen pixels, and their limited discriminative features significantly complicate detection. As depicted in Fig. 1, large-scale satellite video typically exhibits ultrahigh resolutions (e.g., 5000×5000 pixels), where background regions occupy a substantial portion of the frame. Most background pixels are inconsequential for object detection, whereas vehicles predominantly appear along roadways, permitting the selective omission of extensive background areas.

Traditional video-based object detection methods, including background subtraction [2], [10], [11], [12], frame differencing [8], [9], and optical flow [13], [14], [15], rely on motion cues to identify objects. Background subtraction constructs a

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reference model to detect motion changes, frame differencing highlights motion regions through thresholding, and optical flow examines pixel displacement trajectories. However, these techniques struggle with challenges such as sensor motion, lighting variations, and contrast fluctuations, which are particularly problematic in satellite video due to gaze imaging. Conversely, deep learning-based methods [16], [17], [18], [19] have demonstrated superior performance owing to their robust feature extraction capabilities. Anchor-based detection frameworks, such as RetinaNet [20] and Faster R-CNN [16], utilize predefined anchor boxes for object localization but suffer from issues related to sample imbalance and substantial computational overhead. In contrast, anchor-free approaches including CornerNet [21], RepPoints [22], CentripetalNet [23], and CenterNet [19] eliminate the dependence on anchor boxes by directly predicting object boundaries, thereby improving detection efficiency while maintaining high accuracy.

Applying traditional image-based object detection algorithms directly to moving vehicle detection in satellite videos is inefficient, as these methods do not leverage temporal information effectively. Motion-based methods [24], [25] partially address this limitation, but the nonuniform distribution of vehicles in large-scale satellite imagery remains a substantial challenge, adversely impacting detection accuracy and computational efficiency. The sliding window strategy, which segments images into smaller patches, enhances detection [26]. However, its considerable computational demands restrict practical application. To overcome these challenges, recent methods focus on identifying object cluster regions before detection. ClusDet [27] employs clustering networks, whereas DMNet [28] utilizes density maps to isolate clustered regions, effectively reducing background interference. YOLC [29] further refines clustering through the integration of the LSM module, but its requirement for generating a global heatmap significantly increases computational overhead. Despite these advancements, inference speed remains a bottleneck, making the tradeoff between detection accuracy and computational efficiency a persistent challenge in satellite video object detection.

Inspired by these observations, we propose large scenes to clusters (LSCNet), a lightweight and efficient satellite video vehicle detection network. Built on CenterNet [19] for its simplicity, scalability, and high inference speed, LSCNet directly predicts object centers without anchor boxes, thus effectively addressing the challenges associated with densely distributed and small-scale objects. To further enhance performance, we introduce the object cluster module (OCM), an unsupervised, plug-and-play component specifically tailored for satellite videos. Unlike traditional image cropping methods, OCM is memory-efficient, easy to implement, and compatible with spaceborne constraints. By effectively identifying clustered object regions within large-scale scenes, OCM facilitates optimized cropping, enhances detection efficiency, and provides a robust solution for satellite object detection.

To extract richer and more comprehensive vehicle features, LSCNet introduces the multiframe feature aggregation module (MFAM), which exploits correlations across multiple frames. Compared with other complex feature aggregation techniques, MFAM offers a simpler and more efficient design, making it particularly suitable for lightweight architectures and real-time processing scenarios. In addition, we optimize the network's downsampling strategy to enhance small object detection. Moreover, to address small object detection challenges effectively, we refine the regression loss function by integrating Kullback–Leibler divergence (KLD), culminating in the proposed KLD + L_1 loss function. This integrated loss function leverages the advantages of both KLD and L_1 loss. The architecture of the proposed LSCNet is illustrated in Fig. 2, with its primary contributions summarized as follows.

1) To address the object clustering phenomenon in large-scale satellite videos, we propose a lightweight, plug-and-play OCM that adaptively identifies clustered vehicle regions, significantly improving processing efficiency and practical applicability for spaceborne platforms.
2) To overcome limited vehicle feature representation in satellite videos, we introduce a lightweight MFAM. This module efficiently captures spatiotemporal information while maintaining streamlined network architecture, thereby significantly improving vehicle detection performance.
3) To improve small object detection, we develop a regression loss function combining KLD with L_1 loss. This approach effectively enhances the accuracy of small-scale object detection.

The remainder of this article is structured as follows. Section II reviews related work on moving object detection. Section III provides a detailed exposition of the proposed method. Parameter settings and experimental results are presented in Section IV. This article concludes with a summary in Section V.

II. Related Works

This article explores three fundamental aspects of vehicle detection in large-scale satellite videos: feature aggregation in satellite video object detection, small object detection, and focus-and-detect methods. A brief review of the relevant literature is provided in Sections II-A–II-C.

A. Feature Aggregation in Satellite Video Object Detection

In video object detection tasks, feature aggregation [30], [31], [32] is a crucial technique for enhancing detection accuracy and robustness. Its primary goal is to improve object representation by leveraging spatiotemporal information from adjacent frames within video sequences, thereby boosting detection performance. Unlike single-frame image detection, video object detection requires effective exploitation of interframe correlations to address challenges such as object blur, occlusion, and rapid motion in complex scenarios. Traditional video object detection methods commonly employ optical flow [30], [33] or Transformer-based model [34] to capture spatiotemporal dynamics. However, these approaches are often computationally intensive, provide limited attention to small objects, and demand substantial memory and processing resources, particularly for high-resolution satellite videos.

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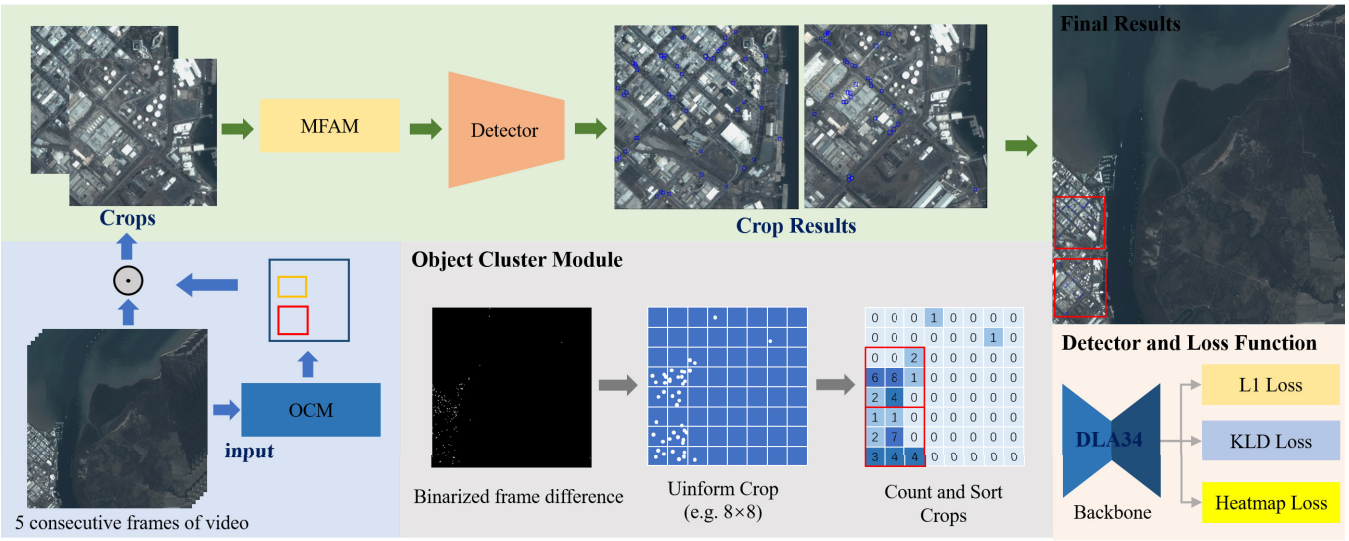


Fig. 2. LSCNet framework integrates the OCM that generates the final detection result by combining the crops with the global image. This framework is specifically tailored for vehicle detection in large-scale satellite videos, ensuring efficient and accurate target recognition. LSCNet primarily consists of three key components: OCM, MFAM, and KLD loss. OCM extracts crops of target-dense regions to enhance detection efficiency, while MFAM and KLD loss collaboratively optimize feature representation and training, thereby improving vehicle representation and overall model performance.

Feature aggregation significantly enhances satellite video object detection by integrating spatiotemporal information from multiple frames. Various techniques have been developed to leverage temporal correlations. Zhou and Li [35] employed unsupervised dense optical flow extraction to capture motion features, while DSFNet [25] introduced a dynamic-static fusion network that integrates 2-D and 3-D backbones to extract both static context and dynamic motion features. ClusterNet [36] adopted a two-stage framework utilizing 2-D convolution on stacked multiframe data, significantly enhancing small object detection. CKDNet [24] improved vehicle keypoint detection by exploiting cross-frame complementary information and incorporating object size estimation. Recent advancements have further refined feature representation. Pi et al. [37] combined motion cues from adjacent frames with Transformer [38] to enhance vehicle feature encoding. SDANet [39] guided networks to focus on road areas while enhancing representation through spatiotemporal embedding. PTFA [40] developed a patchwise temporal–spatial aggregation network modeling semantic interactions. YOLO-HFDM [41] employed historical frame differences to extract effective vehicle features.

Existing algorithms, however, integrate extracted features in ways that risk losing critical information for small vehicles. Furthermore, their reliance on complex network architectures results in significant computational overhead, restricting practical deployment. To improve efficiency while maintaining detection accuracy, future research should develop lightweight aggregation models and explore end-to-end optimization strategies.

B. Small Object Detection

Small object detection faces unique technical challenges and offers broad application potential compared to conventional detection tasks. It must address issues such as information loss, noisy feature representation, low tolerance for bounding box perturbation, and limited training samples [42]. The feature representation quality of small objects is inherently constrained by their limited size and the current limitations of feature extraction methods. Cheng et al. [42] systematically reviewed small object detection methods, categorizing them into six groups: sample-oriented [43], [44], [45], attention-based [46], [47], feature-imitation [48], [49], context-modeling [50], scale-aware [51], and focus-and-detect [27], [28], [29], [52], [53], [54].

For detection optimization, Xu et al. [44] proposed DotD, an alternative to IoU that measures normalized Euclidean distances between bounding box centers, refining matching strategies for small objects. Similarly, RFLA [45] improved detector performance by aligning Gaussian receptive fields with ground-truth labels. KB-RANN [55] employs long- and short-term attention mechanisms to emphasize key regional features, enhancing small object detection. SCRDet [56] integrates pixel- and channel-level attention mechanisms, effectively suppressing background noise. In addition, FCOS [57], an anchor-free detector, incorporates horizontal attention to optimize feature learning, while FBR-Net [58] balances feature representations across pyramid levels.

In satellite video analysis, small object detection presents considerable challenges, primarily due to the high spatial density of small vehicles and the substantial computational burden caused by complex backgrounds. Focus detection, which leverages image cropping-based strategies, enhances memory efficiency while improving detection accuracy, making it particularly effective for high-resolution satellite videos.

C. Focus-and-Detect Methods

Small objects in satellite videos are often nonuniformly distributed, making traditional segmentation-based detection inefficient due to excessive computation in background

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regions. Filtering out nonobject areas substantially reduces redundant calculations, optimizing the detection process and improving inference efficiency. This approach restructures the traditional high-resolution image processing pipeline, mitigating information loss while enhancing feature representation quality.

Yang et al. proposed ClusDet [27], a distributed detection network that exploits semantic relationships and spatial features to enhance detection performance through chip clustering. Building on this, CDMNet [52] and DMNet [28] utilized pixel-level supervision for density estimation, generating precise density maps to characterize object distributions. CRENet [59] employed object search efficiency with an adaptive clustering algorithm, while PRDet [53] incorporated a context learning module into traditional region extraction methods to enhance small object recognition in complex backgrounds. The focus and detect (F&D) [54] framework first detected candidate regions using a focusing network, subsequently enhancing resolution through cropping. YOLC [29] employed an unsupervised LSM module for object localization, but its reliance on global heatmaps increases computational overhead. To address these challenges, we propose the OCM, a general-purpose component designed to enhance computational efficiency while maintaining high detection accuracy. Seamlessly integrable into various satellite video object detectors, OCM provides a simple yet effective optimization strategy for large-scale satellite video analysis.

III. Proposed Method

This article introduces LSCNet, a framework for vehicle detection in large-scale satellite videos, as illustrated in Fig. 2. The architecture integrates the OCM for cropping object-dense regions, significantly improving detection efficiency. Subsequently, the MFAM enhances vehicle feature representations, refining localization accuracy. Finally, the incorporation of Kullback–Leibler Divergence (KLD) into the loss function enhances the training process for small objects, thereby further improving detection performance.

A. Overview of LSCNet

Our proposed LSCNet is an efficient and lightweight vehicle detection model specifically designed for satellite videos. It identifies clustered regions of moving vehicles within large-scale satellite scenes and effectively performs cluster-based detection. The model incorporates an unsupervised, plug-and-play OCM, a lightweight and effective MFAM, and an optimized loss function tailored for small vehicle detection.

The proposed LSCNet builds upon CenterNet [19] by incorporating the MFAM, which aggregates spatiotemporal features to enhance vehicle representation. Traditional object detection methods in satellite videos are inefficient due to the uneven distribution of objects, resulting in substantial computational waste and diminished accuracy. To address this challenge, OCM adaptively identifies and extracts object-dense regions, thereby reducing redundancy and significantly improving detection efficiency. In the detection pipeline, OCM initially isolates key regions containing vehicles, which are subsequently analyzed. High-confidence detections are mapped back to their original positions to refine the final output. However, due to the small proportion of vehicles in satellite images, traditional methods struggle to effectively capture object features, making small object detection particularly challenging. To overcome this limitation, we propose a hybrid loss function that combines KLD loss and L_1 loss. In addition, we expand the main network's input channels from the original 3–15 and optimize the backbone network's downsampling strategy. These improvements enable LSCNet to generate high-resolution heatmap, significantly improving small object detection accuracy. The following sections provide a detailed discussion of the design and implementation of each LSCNet module.

B. Object Cluster Module

Algorithm 1 OCM Algorithm

Input: Binarized frame difference D, the number of grids G_w, G_h, the number of crops n
Output: Bounding boxes of cluster crops $\mathcal{B} = \{\mathcal{B}_i\}_{i=1}^{n}$

$W, H = D.width, D.height$
$w = \frac{W}{Gw}, h = \frac{H}{Gh}$
$M = \text{zeros}(G_w, G_h)$
for $i = 0$ **to** G_w **do**
  for $j = 0$ **to** G_h **do**
    $M[i, j] = \text{sum}(D[w \cdot i : w \cdot (i+1), h \cdot j : h \cdot (j+1)])$
  end
end
$Array \leftarrow \text{topK}(M)$
$\mathcal{B} \leftarrow \text{eight-neighbor}(Array)$
$\mathcal{B} \leftarrow \text{sort}(\mathcal{B})$
if $len(\mathcal{B}) > n$ **then**
  $\mathcal{B} \leftarrow \mathcal{B}[1 \ldots n]$
end
Return $\mathcal{B}$

Efficient region-of-interest (ROI) extraction is crucial for enhancing detection performance, particularly in cropping-based methods. In satellite videos, vehicles are predominantly located along roads and bridges, while background regions constitute the majority of the image, leading to extensive computational waste. Since vehicles typically account for less than 0.01% [60] of the total image, global detection is highly inefficient, especially for high-resolution satellite imagery, where nontarget regions further amplify computational costs. Existing cropping-based methods, such as ClusterNet [36] and F&D [54], employ a two-stage network to localize object-dense areas and optimize resource allocation. However, these methods introduce additional computational overhead, reduce inference speed, and increase model complexity, thus restricting real-time applicability and large-scale deployment. Balancing computational efficiency with detection accuracy remains a significant challenge in satellite video object detection.

To resolve these challenges, this article introduces the OCM, specifically designed to efficiently and accurately detect

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clustered regions in satellite videos. Given the static background and sparse moving objects characteristic of satellite imagery, OCM leverages multiframe differencing to isolate moving pixels and identify moving objects. This intuitive approach effectively exploits the inherent temporal information available in satellite video sequences. Nevertheless, traditional detection methods frequently produce false positives (FPs) and miss faint targets, hindering accurate cluster identification. To mitigate these issues, we incorporate SORT [61] to extract coherent motion trajectories from multiframe difference results. SORT employs a Kalman filter with a constant velocity model, which aligns well with the typical smooth motion of vehicles in satellite videos. By leveraging motion prediction and frame-to-frame association via the Hungarian algorithm, SORT enables reliable tracking without the need for manual threshold tuning. This allows us to implicitly constrain trajectory continuity and velocity, thereby reducing false alarms from noise or nonvehicle motions.

Compared to existing global heatmap-based methods, OCM offers enhanced flexibility, computational efficiency, and suitability for real-time applications. In addition, OCM identifies significant object clusters by ranking pixel densities within grids, effectively selecting the top-K densest regions. Given the multiple object clusters often present in satellite videos, this feature considerably improves detection accuracy. Furthermore, OCM dynamically adapts to varying scenarios, minimizing redundant computations from irrelevant regions. Detailed descriptions and analyses of OCM's implementation are provided in the following sections, with its structure illustrated in Fig. 2.

Candidate region generation is a crucial component of OCM. Initially, a binary mask identifies potential target locations by leveraging satellite videos' multiframe characteristics. Multiframe differencing detects moving pixels by comparing consecutive frames and removes false alarms via SORT [61]. The resultant frame difference image is binarized using an empirical threshold, creating a location mask highlighting probable target regions. Subsequently, this binary heatmap is segmented into smaller grids (e.g., 32×32 pixels), with the top-K densest regions selected as candidate areas. As objects may span adjacent regions, straightforward grid-based segmentation could result in information loss. Following the approach in [28], adjacent densely populated grids are merged using eight-neighbor connectivity to form larger candidate regions. Ultimately, these dense regions are then extracted from the original image to generate k image patches. This overall process is outlined in Algorithm 1.

To enhance detection speed and accuracy, our primary goal is to reduce the number of candidate regions generated. Specifically, for each image, we set $k = 2$ and prioritize larger and denser regions, maximizing detection efficiency. Furthermore, due to its simple and intuitive structure, OCM can be seamlessly integrated into existing object detection frameworks without significant computational overhead. The proposed unsupervised OCM module is widely applicable to various domains, including satellite video analysis and general video surveillance.

C. Multiframe Feature Aggregation Module

To address the limited feature representation of vehicles in satellite videos, several feature aggregation methods have been proposed. Fig. 3 illustrates three typical approaches for aggregating spatiotemporal features. (a) employs independent modules to extract temporal features, which are subsequently fused with spatial features to enhance representation (e.g., DSFNet [25] utilizes a two-stream network, while Zhou and Liu [35] apply dense optical flow for temporal feature extraction). (b) extracts spatial features from multiple frames and fuses them across different time steps. Transformer architectures [38] are widely used in this approach to optimize feature transfer, as demonstrated in TransVOD [34] and the method proposed by Pi et al. [37]. (c) aggregates temporal information at the input level before processing by the backbone network, exemplified by CKDNet [24], YOLO-HFDM [41], and ClusterNet [36]. While (a) effectively integrates temporal and spatial information but introduces additional computational overhead and may negatively affect training stability. (b) risks redundant computations during feature fusion, potentially causing the backbone network to overlook crucial details of small objects. In contrast, (c) achieves a balance between feature extraction efficiency and model complexity, enhancing detection performance while maintaining a lightweight architecture.

Building on this analysis, this article proposes an efficient and lightweight feature aggregation method inspired by spatial attention mechanisms and ClusterNet [36]. The proposed MFAM features a simple yet effective design, enabling the utilization of adjacent frames to enhance vehicle feature extraction in the current frame. In addition, MFAM maintains low computational cost, providing a practical and efficient solution for feature aggregation. The following section provides a detailed explanation of MFAM.

Fig. 4 illustrates the MFAM workflow. This module leverages multiframe information to enhance vehicle feature extraction. Initially, five consecutive image frames are concatenated along the channel dimension, forming a multiframe feature M with dimensions $W \times H \times 15$, which serves as the basis for subsequent feature association. Next, conv, batch normalization, and sigmoid-weighted linear unit (SiLU) activation (collectively referred to as CBS) operations are applied to M, resulting in feature P. Subsequently, CBS and the sigmoid function are applied to P, which is then multiplied and summed with itself to produce feature $Q = P[1 + \text{sigmoid}(\text{CBS}(P))]$. This formulation can be interpreted as a soft, learnable attention mechanism that adaptively emphasizes important features while preserving residual information. Unlike explicit attention modules that compute weighted maps across spatial or channel dimensions, this design offers a lightweight and implicit attention behavior, enabling the network to amplify interframe salient regions and suppress redundant background noise with minimal overhead. Finally, Q undergoes two additional CBS operations to yield the final fused feature F with 16 channels

$$\text{CBS} = \text{SiLU}(\text{BN}(\text{con}(\cdot))) \tag{1}$$

$$P = \text{CBS}(M) \tag{2}$$

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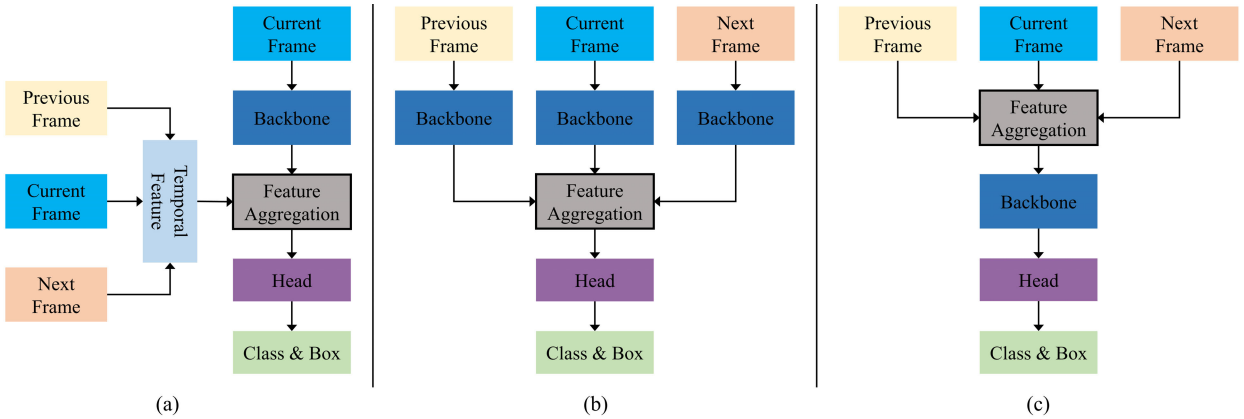


Fig. 3. Several feature aggregation frameworks have been proposed for satellite video processing. (a) Some methods [25], [35] extract temporal features using optical flow or dual-stream networks, which are then fused with spatial features to improve detection accuracy and efficiency. (b) Other approaches [34], [37] first extract features from multiple frames and aggregate them, often utilizing Transformers for the fusion process. (c) Proposed method performs feature aggregation prior to passing the aggregated features into the backbone network (e.g., CKDNet [24], YOLO-HFDM [41], and ClusterNet [36]). While methods (a) and (b) tend to increase both network complexity and computational load, the proposed framework improves the extraction of small target features, enhancing detection efficiency and reducing overall complexity.

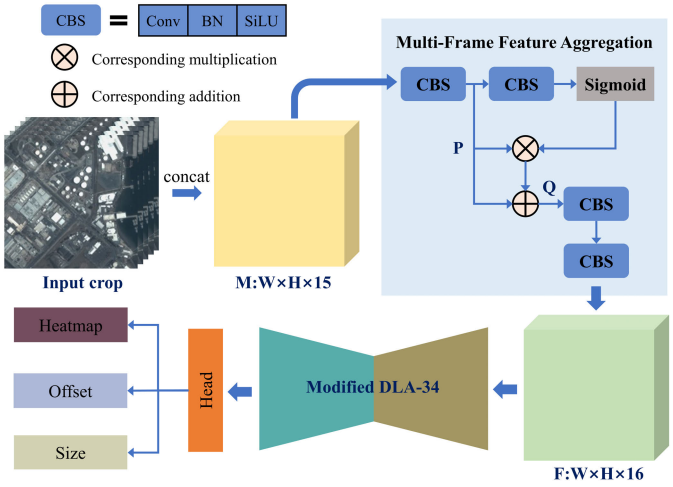


Fig. 4. Architecture of MFAG.

$$Q = P[1 + \text{sigmoid}(\text{CBS}(P))] \tag{3}$$

$$F = \text{CBS}(\text{CBS}(Q)) \tag{4}$$

where conv in CBS is the 2-D convolution with kernel = 3, stride = 1, and padding = 1. Sigmoid represents the sigmoid function.

In contrast to ClusterNet [36], which directly stacks multiple input frames for feature aggregation, MFAM enhances inter-frame feature correlations through residual structures and attention mechanisms. Its primary objective is to amplify subtle vehicle features and preserve essential information while minimizing redundant data.

Small vehicles in multiframe images exhibit stronger correlation information compared to single-frame images. Unlike methods aggregating features after backbone processing, MFAM integrates multiframe information prior to feature extraction, reducing information loss and enhancing small target representation. Through loss function backpropagation and gradient descent optimization, MFAM effectively captures and integrates key features of small vehicles across multiple frames.

D. Loss Function

The loss function of CenterNet [19] is primarily composed of three components

$$L_{\text{det}} = L_k + \lambda_{\text{size}} L_{\text{size}} + \lambda_{\text{off}} L_{\text{off}} \tag{5}$$

where L_k represents the focal loss used in CenterNet [19], L_{off} denotes the center point offset loss, and L_{size} refers to the size regression loss. L_{off} and L_{size} quantify the similarity between bounding boxes and are both optimized using L_1 loss.

CenterNet [19] utilizes the L_1 loss for size and center point regression. However, this approach has notable limitations. First, it independently regresses the height and width of an object, disregarding their inherent interdependence. In addition, the uniform distribution characteristic of L_1 loss limits precise size estimation for small objects, as highlighted by RFLA [45] and DotD [44]. Moreover, the L_1 loss can create inconsistencies between regression accuracy and actual detection performance, meaning that reduced training loss does not necessarily correlate with improved detection outcomes. Last, even with comparable L_1 losses, significant discrepancies can persist between predicted and ground-truth bounding box IoUs, further emphasizing this method's limitations in size regression.

To mitigate these challenges, we propose employing the KLD [45] to evaluate bounding box similarity. Specifically, a ground-truth (GT) box is represented as $B(x, y, w, h)$, where (x, y) represents the center point, and w and h denote width and height, respectively. The GT box is modeled as a standard 2-D Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with the center point as the mean vector and the covariance matrix defined using the squares of half side lengths

$$\boldsymbol{\mu} = \begin{bmatrix} x \\ y \end{bmatrix}, \boldsymbol{\Sigma} = \begin{bmatrix} \frac{w^2}{4} & 0 \\ 0 & \frac{h^2}{4} \end{bmatrix}. \tag{6}$$

KLD is a fundamental statistical measure quantifying the difference between two probability distributions. For

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two Gaussian distributions $\mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ and $\mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$, the closed-form solution for KLD is calculated as follows:

$$D_{\mathrm{kl}} = \frac{1}{2}\left(\mathrm{tr}\left(\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Sigma}_2\right) + \ln\frac{|\boldsymbol{\Sigma}_1|}{|\boldsymbol{\Sigma}_2|} + (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^{\top}\boldsymbol{\Sigma}_1^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) - 2\right). \tag{7}$$

Equation (7) can be further simplified as

$$D_{\mathrm{kl}} = \frac{w_1^2}{2w_2^2} + \frac{h_1^2}{2h_2^2} + \ln\frac{w_2}{w_1} + \ln\frac{h_2}{w_1} + \frac{2(x_1 - x_2)^2}{w_2^2} + \frac{2(y_1 - y_2)^2}{h_2^2} - 1 \tag{8}$$

where (x_1, y_1, w_1, h_1) and (x_2, y_2, w_2, h_2) represent the coordinates, width, and height of GT and predicted box, respectively. However, solely relying on D_{kl} [(8)] may lead to excessive sensitivity to variations in errors. Therefore, we introduce a nonlinear function to transform D_{kl} into an affinity metric, formulating the final KLD-based loss as follows:

$$L_{\mathrm{kld}} = \frac{1}{1 + e^{-D_{\mathrm{kl}}}} - 1. \tag{9}$$

To improve bounding box regression, we integrate KLD loss with L_1 loss to leverage their complementary strengths. KLD models spatial deviations with Gaussian weighting, making it well-suited for small object localization by emphasizing center alignment and subtle scale differences. In contrast, IoU-based losses rely on box overlap and often fail when predictions do not intersect with GT. Compared to GWD [45], which measures overall distributional similarity, KLD is more responsive to center and scale variations, albeit at the cost of stability when predictions are distant from GT. Combining KLD with L_1 enhances localization accuracy while maintaining numerical stability. The final loss function is defined as

$$L_{\mathrm{det}} = L_k + \lambda_{\mathrm{kld}} L_{\mathrm{kld}} + \lambda_{l1} L_1 \tag{10}$$

where λ_{kld} is set to 1 and λ_{l1} is set to 0.3. The enhanced loss function can significantly improve LSCNet's vehicle detection performance in satellite videos.

IV. Experiments and Analysis

This section presents the datasets, evaluation metrics, and implementation details employed in our experiments. Subsequently, comparative experiments are conducted from both qualitative and quantitative perspectives to demonstrate the proposed method's superiority over existing state-of-the-art approaches. Finally, a comprehensive ablation study is presented to assess the individual contributions of each module and loss function.

A. Datasets

To evaluate the performance of our method, we conduct experiments using two satellite video datasets: VISO [60] and LSSV. VISO [60], a publicly available satellite video vehicle detection dataset, contains diverse scenes with complex backgrounds, making it suitable for assessing object recognition capabilities. The large scene satellite video (LSSV) dataset, newly introduced in this article, is specifically designed for large-scale satellite video object detection. The training set comprises 33 895 images and 815 559 instances, while the test set contains 3206 images and 152 667 instances. As illustrated in Fig. 5, the specifications of datasets are summarized in the following.

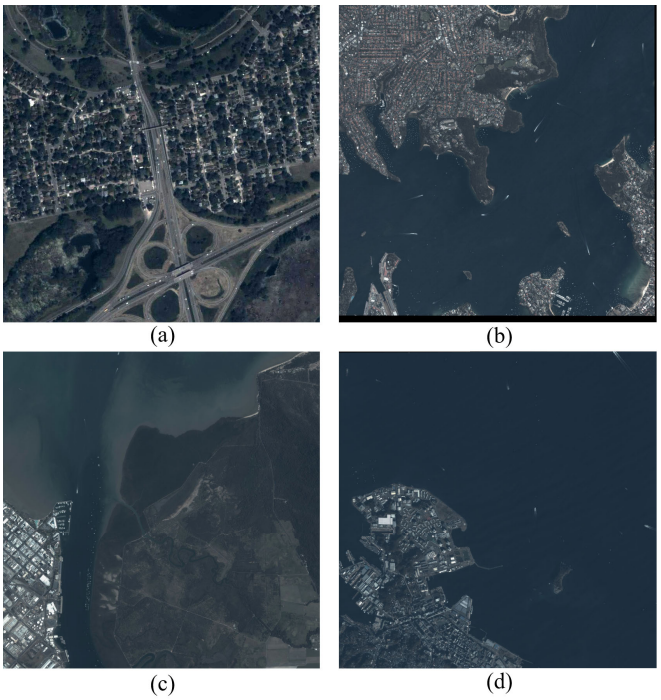


Fig. 5. Example frames from the satellite video dataset. (a) Sample from VISO with a resolution of 1024 × 1024, while (b)–(d) are from LSSV, each with a resolution of 5000 × 5000. These examples illustrate that large-scale scenes primarily consist of irrelevant background regions. We crop smaller regions containing vehicles from the original large-scale data for testing and training.

1) VISO [60]: This dataset is divided into training and test subsets. The training subset consists of 70 video samples with a resolution of 512 × 512 pixels, and the test subset contains six videos with a resolution of 1024 × 1024 pixels.

2) LSSV: The LSSV dataset is captured by the Jilin-1 satellite, featuring footage from Macau, China; Queensland, Australia; and Yokohama, Japan. The original video resolution is 5000 × 5000 pixels. For comparative evaluation, smaller regions containing vehicles are cropped from the original large-scale data. Specifically, we extracted 14 cropped videos (resolution: 512 × 512 pixels) for training and selected four cropped videos (resolution: 1024 × 1024 pixels) for testing. In addition, the LSSV dataset provides large-scale vehicle detection scenarios, enabling the demonstration of LSCNet's rapid detection capabilities.

B. Experimental Setups

LSCNet is implemented using PyTorch 1.7.0 and trained on an Ubuntu 18.04 system with four NVIDIA A800 GPUs and an Intel Xeon Gold 6246 CPU.

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The model is trained for 70 epochs using the Adam [62] optimizer. The initial learning rate is set to 1.25×10^{-4} and reduced by a factor of 0.1 after 30, 45, and 60 epochs. The batch size is set to 32. During training, the number of input frames T is set to 5. The grid size $G_w \times G_h$ is set to 32×32, and the number of crops n is set to 2. In addition, in the loss function, the hyperparameter λ_{kld} is set to 1.

C. Evaluation Metric

To comprehensively evaluate the proposed method, precision, recall, and $F1$-score are adopted as primary evaluation metrics. Specifically,

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{12}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{13}$$

where true positives (TPs), false negatives (FNs), and FPs represent the number of correct detection, missed detection, and false alarms, respectively. Although Intersection over Union (IoU) is commonly used in object detection evaluations, it exhibits limited stability when detecting small targets in satellite videos, where minor variations in bounding boxes can significantly affect IoU values. To mitigate this issue, we adopt a center-point matching strategy following [25], [36]. A predicted bounding box is classified as a TP if its center is within 5 pixels of the corresponding GT box.

In addition, frames per second (FPS) is employed as the speed evaluation metric, calculated using the following formula to assess the model's detection speed:

$$\text{FPS} = \frac{1}{t} \tag{14}$$

where t is the inference time of a single image.

To more concisely evaluate the overall performance of the proposed real-time model, this article introduces a novel evaluation metric that covers detection accuracy, inference speed, and model complexity simultaneously (short for ASC), which is computed as follows:

$$\text{ASC} = \frac{\text{F1} \times \text{FPS}}{\text{Param} \times \text{FR}} \times \text{C} \tag{15}$$

where Param denotes the number of model parameters and FR represents the video frame rate. For the Jilin-1 satellite video, FR is set to 15. C is a constant and is set to 100. The proposed ASC metric incorporates detection accuracy, inference speed, and model complexity to assess the model's overall processing capability in real-world applications. Specifically, the term Param $\times$ FR reflects the overall computational burden under real-time constraints, as a larger model running at higher frame rates requires more processing resources. This formulation encourages efficient models with good accuracy and low complexity. A higher ASC score indicates greater practicality and efficiency in deployment scenarios.

D. Comparison to the State of the Art

To substantiate the effectiveness of LSCNet, we conduct a comparative analysis against representative traditional methods, including GoDec [63], FPCP [64], PSPG [65], and deep learning-based object detection algorithms, such as Deformable DETR [66], CenterNet [19], and DSFNet [25]. LSCNet-Lite denotes a lightweight variant of LSCNet achieved through channel pruning. Detection performance is evaluated using recall, precision, and $F1$ score, with all methods retrained on the same datasets used in this article.

1) Quantitative Results: To systematically evaluate detection performance, we conduct quantitative experiments on the VISO [60] and LSSV datasets. The corresponding results are shown in Tables I and II, respectively. As observed, LSCNet achieves the highest detection accuracy, with average $F1$ scores of 89.1%–89.3% on the VISO [60] and LSSV datasets, respectively, demonstrating substantial improvements over existing methods. Notably, compared with the second-best DSFNet [25], LSCNet achieves performance gains of 2.8% and 1.5%, respectively, further validating its superiority. In addition, LSCNet significantly reduces FP rates, achieving false alarm rates of 11.9%–9.1% on the VISO and LSSV datasets, respectively. These results represent reductions of 5.8%–2.3% compared to DSFNet, demonstrating enhanced robustness in suppressing background noise and nonvehicle interference.

RPCA-based methods such as GoDec [63], FPCP [64], and PSPG [65] exhibit suboptimal performance in scenes with dense vehicle distributions. These traditional approaches depend heavily on background modeling and struggle to distinguish low-contrast vehicles in complex environments, resulting in high missed detection rates. Similarly, image-based methods such as CenterNet [19] and Deformable DETR [66] are limited by their inability to fully exploit interframe temporal information, thereby reducing their effectiveness in capturing vehicle motion characteristics.

Notably, DSFNet [25], which combines static and dynamic features from multiple frames, achieves the second-best detection performance. This result underscores the importance of incorporating spatiotemporal features in satellite video data. However, in comparison, LSCNet still demonstrates a clear advantage in overall detection performance. Its design not only enhances detection accuracy but also exhibits greater robustness in complex scenarios. Its streamlined architecture also makes it well-suited for small object detection.

2) Qualitative Results: To further assess detection robustness in complex scenes, we present the qualitative detection results in Fig. 6. (a)–(g) corresponds to the methods GoDec [63], FPCP [64], PSPG [65], CenterNet [19], Deformable DETR [66], DSFNet [25], and LSCNet, respectively. The first four rows display crop results from the VISO [60] dataset, while the last four rows are from the LSSV dataset. As shown in the comparison, it is evident that our LSCNet achieves superior and more robust detection performance, maintaining a high detection rate and low false alarm rate, whereas other comparison methods exhibit varying degrees of false detections and missed detections.

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TABLE I
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT COMPARISON METHODS ON THE VISO DATASET: THE BEST RESULTS ARE HIGHLIGHTED IN **RED**, THE SECOND BEST IN **BLUE**, AND THE THIRD BEST IN **GREEN**

Method	Video1			Video2			Video3			Video4			Video5			Video6			AVERAGE		
	Re	Pr	F1	Re	Pr	F1	Re	Pr	F1	Re	Pr	F1	Re	Pr	F1	Re	Pr	F1	Re	Pr	F1
GoDec [63]	61.5	36.9	46.1	76.3	22.4	34.7	66.3	24.3	35.5	53.3	30.3	38.7	56.5	40.6	47.2	52.0	15.8	24.2	61.0	28.4	37.7
FPCP [64]	71.8	26.1	38.3	67.5	34.8	45.9	55.3	33.4	41.6	59.9	19.9	29.9	61.7	30.1	40.4	52.7	16.8	25.5	59.8	28.7	38.3
PSPG [65]	66.9	32.0	43.3	63.6	46.1	53.4	61.4	32.7	42.7	56.3	24.5	34.1	58.8	36.9	45.3	46.3	22.0	29.8	58.9	32.3	41.4
Deformable DETR [66]	73.8	18.9	30.7	75.0	29.0	41.8	55.8	36.5	44.2	81.0	12.6	21.8	76.0	13.1	22.3	64.9	28.3	39.4	71.8	23.1	33.3
CenterNet [19]	80.1	69.2	74.3	50.6	83.9	63.2	68.2	82.3	74.6	61.6	71.2	66.1	61.7	64.3	63.0	63.1	84.0	72.1	64.2	75.8	68.9
DSFNet [25]	**90.9**	**71.8**	**80.3**	**88.2**	**91.6**	**89.9**	**91.0**	**83.8**	**87.3**	**92.2**	**85.7**	**88.9**	**94.3**	**85.2**	**89.5**	**89.5**	**75.5**	**81.9**	**91.0**	**82.3**	**86.3**
LSCNet-Lite	**90.8**	**82.2**	**86.3**	**78.9**	**92.6**	**85.2**	**87.5**	**84.7**	**86.1**	**90.9**	**83.0**	**86.7**	**91.3**	**85.4**	**88.3**	**91.2**	**82.3**	**86.5**	**88.4**	**85.0**	**86.5**
LSCNet (Ours)	**91.6**	**84.2**	**87.8**	**85.1**	**92.5**	**88.7**	**88.5**	**87.4**	**87.9**	**92.8**	**88.2**	**90.4**	**91.3**	**88.8**	**90.1**	**92.0**	**87.4**	**89.7**	**90.2**	**88.1**	**89.1**

TABLE II
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT COMPARISON METHODS ON THE LSSV DATASET: THE BEST RESULTS ARE HIGHLIGHTED IN **RED**, THE SECOND BEST IN **BLUE**, AND THE THIRD BEST IN **GREEN**

Method	Video1			Video2			Video3			Video4			AVERAGE		
	Re	Pr	F1	Re	Pr	F1	Re	Pr	F1	Re	Pr	F1	Re	Pr	F1
GoDec [63]	37.5	9.4	15.0	31.3	4.6	8.0	22.1	1.9	3.5	4.6	5.2	4.9	23.2	5.2	7.7
FPCP [64]	7.5	15.2	10.1	4.2	7.5	5.4	15.2	3.2	5.3	6.9	4.9	5.7	8.4	7.7	6.6
PSPG [65]	22.9	11.3	15.1	13.1	4.5	6.8	24.0	1.2	2.4	13.5	4.0	6.2	16.8	5.2	7.5
Deformable DETR [66]	43.5	41.7	42.6	77.9	15.5	25.9	82.2	10.5	18.6	76.7	12.2	21.0	70.1	20.0	27.0
CenterNet [19]	14.4	61.7	29.5	28.0	20.0	23.3	35.0	22.3	27.2	22.9	31.5	26.5	24.2	29.2	22.0
DSFNet [25]	**84.2**	**92.3**	**88.1**	**89.9**	**89.7**	**89.8**	**92.8**	**87.7**	**90.2**	**81.4**	**84.6**	**83.0**	**87.1**	**88.6**	**87.8**
LSCNet-Lite	**83.6**	**88.9**	**86.2**	**87.5**	**86.4**	**87.0**	**91.8**	**83.6**	**87.5**	**81.1**	**79.3**	**80.2**	**86.0**	**84.6**	**85.2**
LSCNet (Ours)	**84.2**	**92.6**	**88.2**	**88.0**	**94.0**	**90.9**	**94.6**	**92.2**	**93.4**	**84.4**	**84.7**	**84.5**	**87.8**	**90.9**	**89.3**

Specifically, traditional background modeling-based methods fail to detect low-contrast vehicles effectively and are sensitive to background noise, lighting, and shading variations. As a result, they suffer from poor stability. Similarly, CenterNet [19] and Deformable DETR [66], which rely on single-frame detection, cannot effectively leverage temporal context, often misclassifying static and dynamic vehicles.

In contrast, DSFNet [25] improves detection accuracy by integrating interframe information. However, for vehicles with blurred texture features, this method still suffers from false alarms and missed detections, suggesting that its adaptability to complex backgrounds remains an area for improvement. Finally, our LSCNet demonstrates strong robustness across various environments, accurately detecting moving vehicles and maintaining reliable performance under complex backgrounds.

E. Time Efficiency Analysis

In real-world applications, the real-time performance of detection algorithms is of considerable importance. To this end, we systematically evaluated the detection speed, inference time, and parameter scale of each model, with the results summarized in Table III. All evaluations are conducted on input images with a resolution of 1024×1024. Among RPCA-based methods, FPCP [64] demonstrates shorter inference time and faster detection speed. However, its detection performance remains limited. In contrast, PSPG [65] exhibits relatively low inference efficiency due to the high computational complexity involved in its background modeling process.

TABLE III
PARAMETERS AND SPEED OF DIFFERENT COMPARISON METHODS

Methods	F1(%)	Time(ms)	FPS	Params(M)	ASC
GoDec [63]	37.7	252.5	3.9	-	-
FPCP [64]	38.3	**20.8**	**48.1**	-	-
PSPG [65]	41.4	1234.5	0.8	-	-
Deformable DETR [66]	33.3	121.5	8.2	38.31	0.47
CenterNet [19]	68.9	**46.3**	**21.6**	16.40	**6.05**
DSFNet [25]	**87.8**	286.4	3.5	**6.71**	3.05
LSCNet-Lite	**86.5**	**47.5**	**21.1**	**2.47**	**49.26**
LSCNet (Ours)	**88.6**	152.3	6.6	**6.63**	**5.88**

Regarding deep learning-based methods, CenterNet [19] and Deformable DETR [66] operate on a single frame, while DSFNet [25] and our LSCNet utilize multiple frames as input to fully exploit temporal information. Furthermore, LSCNet-Lite, derived via channel pruning, offers a lightweight solution. Although CenterNet [19] supports real-time inference, it lacks sufficient detection accuracy. In comparison, LSCNet-Lite achieves a processing speed of 21.1 FPS while maintaining high detection performance, with an $F1$ score of 86.5%, effectively balancing real-time capability and accuracy. It is worth noting that deformable DETR [66] suffers from heavy parameters due to Transformer [38] architecture,

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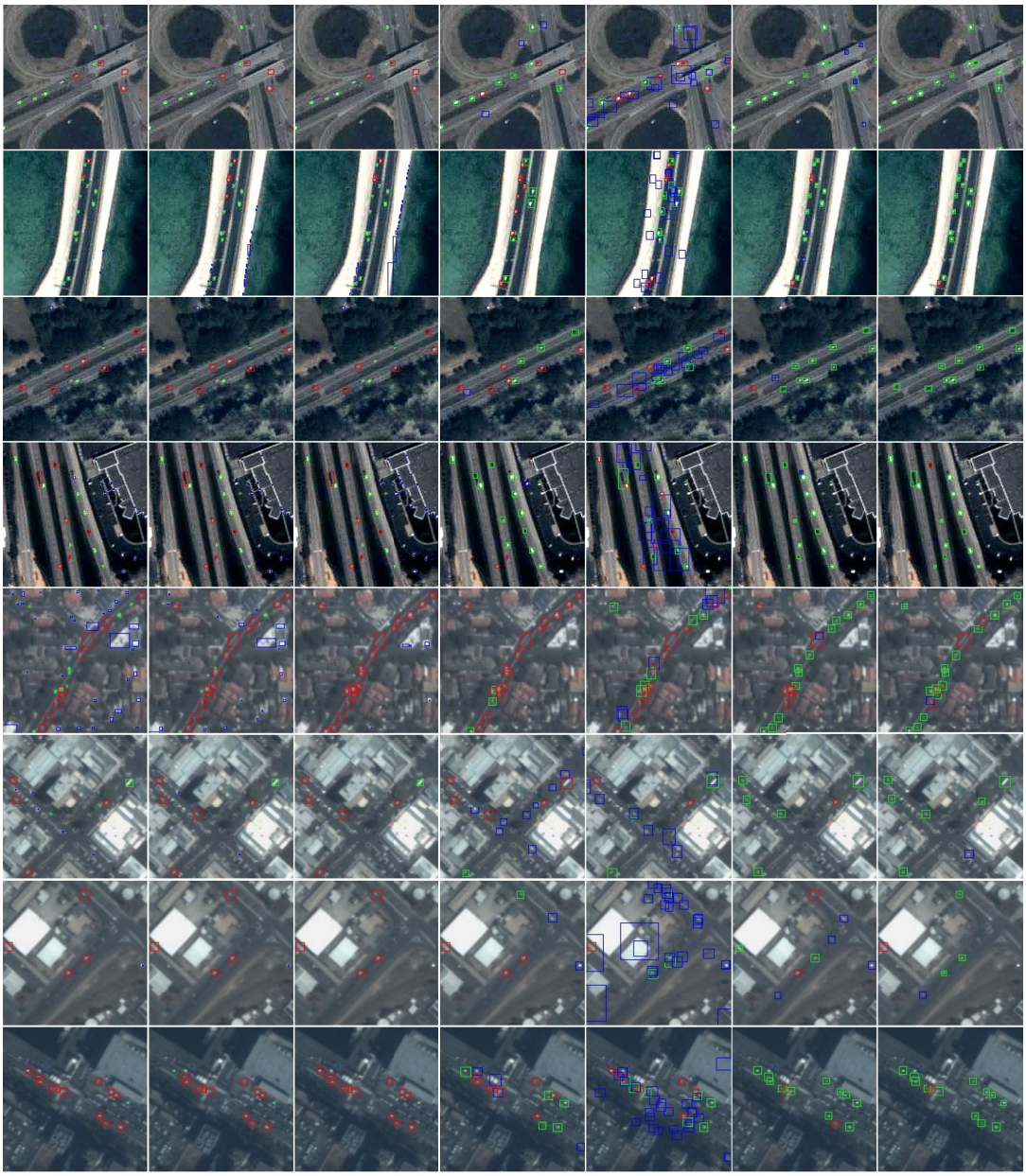


Fig. 6. Qualitative results of different comparison methods. The first four rows of scenes are from VISO [60], and the last four rows of scenes are from LSSV. (a) GoDec [63]. (b) FPCP [64]. (c) PSPG [65]. (d) CenterNet [19]. (e) Deformable DETR [66]. (f) DSFNet [25]. (g) LSCNet.

which adversely affects inference efficiency. By contrast, LSCNet-Lite has only 2.47 M parameters, offering a more lightweight design. Notably, LSCNet-Lite achieves an ASC score of up to 49.26, substantially outperforming all competing methods. Among all evaluated methods, LSCNet achieves an optimal balance between detection accuracy and processing speed, thereby satisfying the stringent real-time requirements of moving vehicle detection in satellite video applications.

F. Large Scene Detection Results

Our LSCNet is a novel detection network specifically designed for large-scale satellite video vehicle detection. In contrast to most existing approaches, which rely on pre-cropped small image patches, LSCNet enables end-to-end analysis directly on high-resolution satellite video frames, improving adaptability and simplifying the processing pipeline. We conduct an evaluation of the model on three Jilin-1 satellite videos with a resolution of 5000 $\times$ 5000, and the detection results are presented in Fig. 7. Each row in Fig. 7 presents the detection results for one video. In the first case, one object cluster region is detected and enlarged for visualization. In the second and third cases, two clusters are detected and visualized accordingly. Overall, the results demonstrate that LSCNet is capable of rapid and accurate detection of high-resolution satellite videos.

As illustrated in Fig. 7, the original satellite images contain a substantial number of invalid regions. With the integrated OCM module, LSCNet quickly identifies and focuses on vehicle-dense regions, avoiding unnecessary computation.

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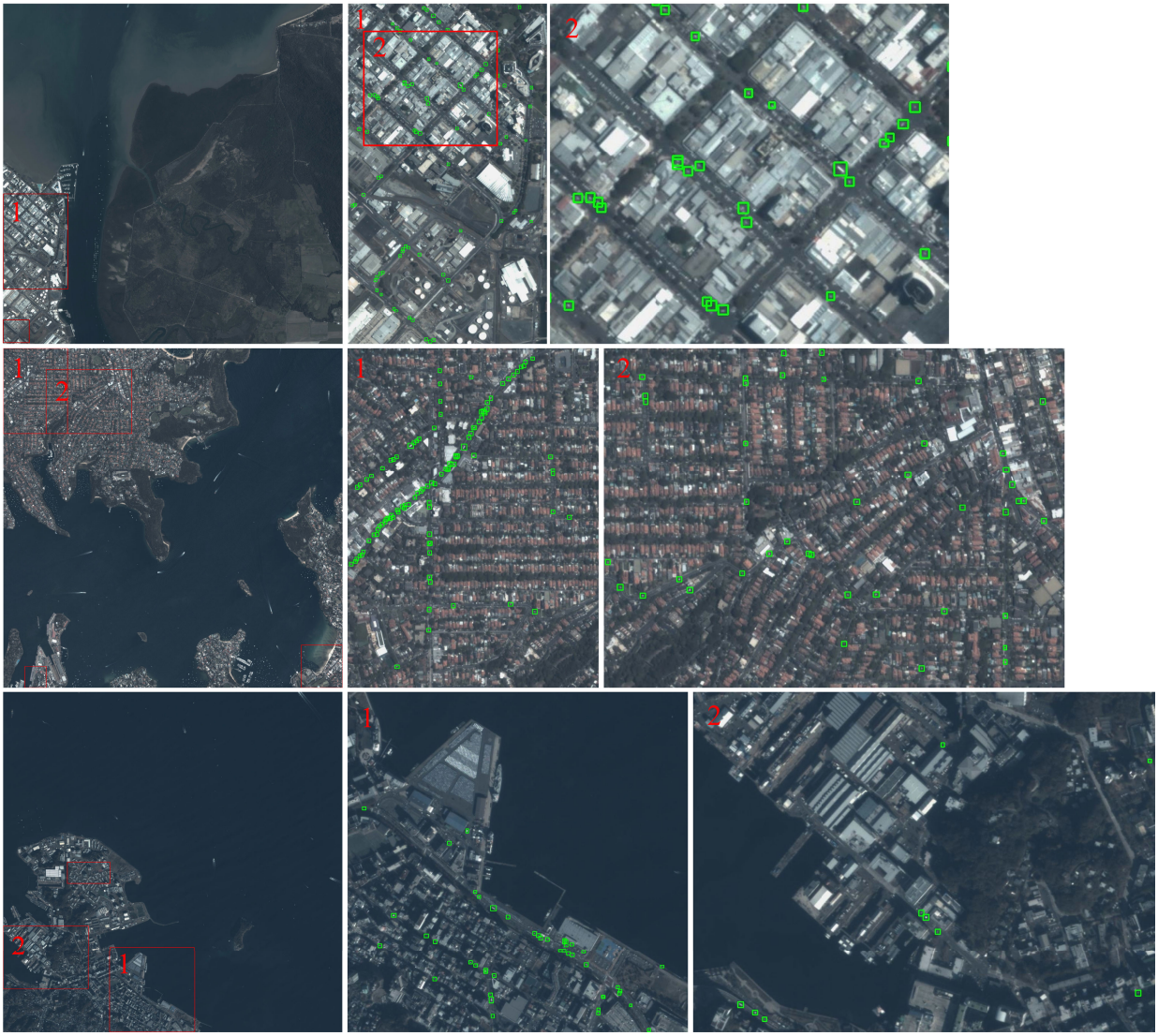


Fig. 7. Large-scale scene detection results on three 5000 × 5000 Jilin-1 satellite videos. Each row corresponds to the detection results from one video. In the first sample, the model identified only a single object cluster region, which is enlarged to provide a clearer view of the detection performance. In the second and third samples, the model detected two object cluster regions in each case and generated the corresponding detection result visualizations.

Subsequent detection is then carried out within these focused regions. The overall time consumption is reported in Table IV. The overall processing time of LSCNet is only 0.57 s. Compared with traditional sliding window or patch-based strategies, the proposed focus-and-detect approach improves efficiency and reduces computational overhead, making it ideal for large-scale satellite video applications.

Furthermore, this method demonstrates strong applicability to resource-constrained satellite platforms. By eliminating extensive redundant computation, it conserves valuable onboard computing resources, making it highly promising for deployment in embedded satellite systems.

G. Ablation Study

This section presents ablation experiments to evaluate the contributions of the MFAM module and the proposed loss function to LSCNet's performance. The experimental results confirm that both components significantly improve detection accuracy.

1) Effectiveness of MFAM: To evaluate the effectiveness of the MFAM and the impact of varying input frame numbers on detection performance, we conducted multiple sets of comparative experiments. The corresponding quantitative results are summarized in Table V. As illustrated by the results, incorporating MFAM yields significant improvements in recall, precision, and $F1$ score, with relative increases of 6.5%, 4.8%, and 6.0%, respectively. The baseline model operates on single-frame satellite images, which inherently fail to leverage temporal continuity across video sequences. This limitation compromises its ability to accurately differentiate between stationary and moving vehicles, especially in dynamic environments. Furthermore, single-frame inputs are more susceptible to common visual artifacts such as motion blur, occlusion, and low contrast, which can severely impair detection accuracy.

By comparing the performance of models using different numbers of input frames, we observe that increasing the number of frames enhances detection accuracy, albeit at the cost of increased computational latency. Considering the tradeoff

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TABLE IV
LARGE SCENE PROCESSING TIME OF DIFFERENT COMPARISON METHODS

Methods	video1	video2	video3	Average
Deformable DETR [66]	3.17	3.23	3.19	**3.20**
CenterNet [19]	1.16	1.24	1.17	**1.19**
DSFNet [25]	7.30	7.31	7.34	7.31
LSCNet (Ours)	0.55	0.60	0.56	**0.57**

TABLE V
PERFORMANCE COMPARISON WITH AND WITHOUT MFAM UNDER DIFFERENT FRAME SETTINGS

MFAM	Frames(n)	Re(%)	Pr(%)	F1(%)	Params(M)
Without	-	83.7	83.3	83.1	6.62
With	3	89.0	87.9	88.4	6.63
	5	90.2	88.1	89.1	6.63
	7	90.0	88.5	89.2	6.64

TABLE VI
PERFORMANCE COMPARISON OF DIFFERENCE LOSS ON THE VISO DATASET

Index	Methods	Re(%)	Pr(%)	F1(%)
A	Baseline(L_1)	89.03	88.28	88.66
B	L_1 + GWD	90.08	87.92	88.98
C	L_1 + KLD	90.01	88.32	89.16
D	L_1 + GWD + KLD	90.26	88.16	89.20

between accuracy and efficiency, we ultimately selected $n = 5$ as the optimal frame number setting.

2) Effectiveness of Loss Function: To further enhance detection performance, we refined the original regression loss function. Building upon the traditional L_1 loss, we introduced KLD loss, which is particularly well-suited for small object localization. For comparative purposes, we also incorporated GWD [45] loss. The weighting coefficients λ_{kld} and λ_{gwd} for KLD and GWD are set to 1 and 0.15, respectively. The corresponding experimental results are summarized in Table VI. As indicated by the results, the model's $F1$ score increased by 0.6% following the adoption of KLD loss, demonstrating its clear advantages for small object detection. A detailed analysis reveals that the combined application of KLD losses significantly enhances the model's ability to detect and localize small-scale targets.

V. CONCLUSION

In this article, we present a novel framework, LSCNet, for large-scale satellite video vehicle detection. LSCNet is designed to tackle key challenges such as high image resolution, wide-area scene coverage, and sparse object distribution in satellite videos. It integrates a plug-and-play OCM to efficiently identify vehicle cluster regions, significantly reducing computational overhead and improving processing efficiency. LSCNet also captures spatiotemporal features across multiple frames using the MFAM, which enhances the representation of moving vehicles and addresses the limitations of single-frame detection in cases of low texture quality. To improve small object detection, it reduces downsampling operations and retains high-resolution feature maps. In addition, the use of KLD loss helps refine localization accuracy for small objects. Experimental results show that LSCNet achieves state-of-the-art accuracy in complex scenarios while maintaining real-time performance. However, its current multiframe feature aggregation still has limitations in modeling temporal dependencies. Future work may explore more effective interframe modeling strategies to further improve the detection of moving vehicles. Moreover, we plan to release the LSSV dataset to the public in order to support further research in satellite video object detection.

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